

Fundamentals of Inference

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Agenda for Today

- Course content overview
- **Module 7:** An Overview of Different Methods for Conducting Statistical Inference (primarily with survey data)

Readings

- Available via Canvas
- In general, these are **foundational** articles
- Some are difficult...
 - Read get sense of the ideas
 - Discuss with other students
 - Ask questions in class

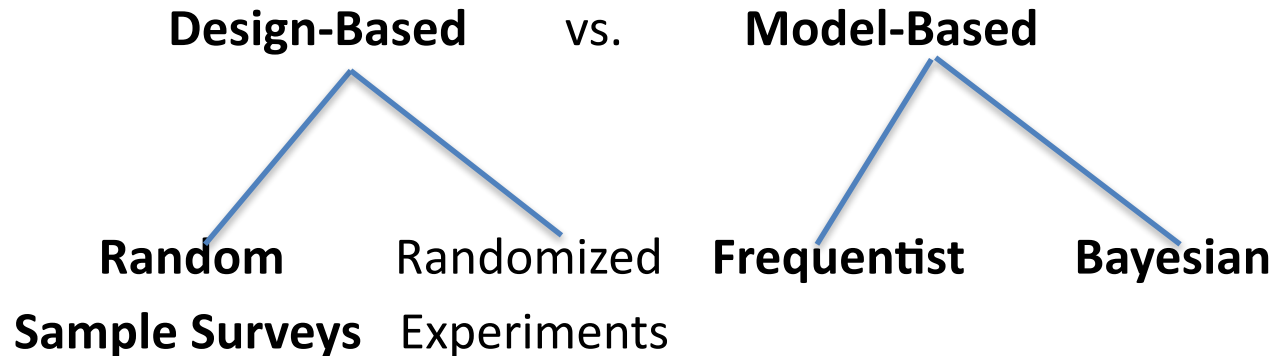
Grading

- Homework:
 - Assignments **available** via Canvas
 - Please **submit** via Canvas

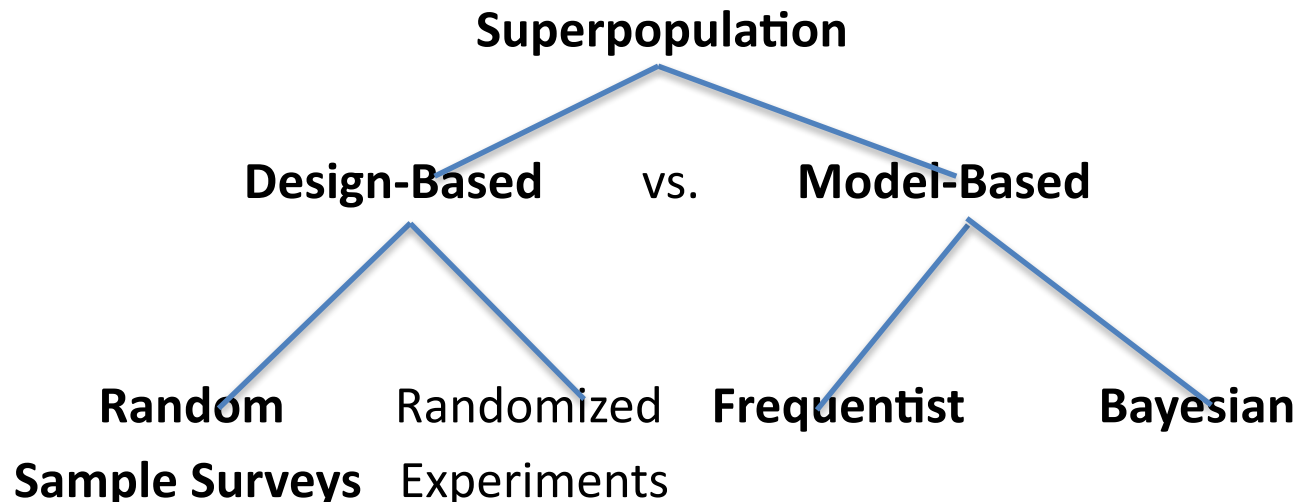
OVERVIEW

Topics in Inference

- **Methods for Making Finite Population Inference**



- **Add Superpopulation Model to Finite Population**



Overview: Modes of Inference

- Each method makes different assumptions
 - **Design-based** inference (**nonparametric**)
 - Adding superpopulation inference (**model-based**)
 - Frequentist and Bayesian finite population inference (**model-based**)
- These differences are important to understand and appreciate conceptually

Overview: Use of Auxiliary Data

- How do we make use of information that we have about the population or full sample?
- How can we improve statistical inference using auxiliary data?
- Covered in Weeks 10-12

METHODS OF INFERENCE

Basic Concepts: Agenda

- Inference
- Inductive vs. Deductive Reasoning
- Design-Based vs. Model-Based Approaches

Basic Concepts: Inference

- **Inference (one definition):**

“The process of deducing properties of the underlying distribution by analysis of data.”

– A Dictionary of Statistics, Upton and Cook, 2008

- Usually accomplished using a sample from a population (Finite or Infinite (superpopulation) or Both)
- ***What is an “underlying distribution”?***

Basic Concepts: Inference

- **Inference:**

“Statistical inference is the process of drawing conclusions about an underlying population based on a sample or subset of the data.”

- An Introduction to Environmental Forensics,
Gauthier and Hawley, 2015

- ***What is the difference between a “sample” and a “subset”? Is a sample a type of subset?***

Basic Concepts: Inductive/Deductive

- Goal of any statistical analysis: Infer, including estimating, about a quantity of interest (usually a **unknown parameter**, such as a proportion or a regression parameter) based on a sample of observations
- Two reasoning or logical approaches:
 - **Inductive** reasoning: Use the particular (sample) and draw conclusions about the general population (**Bayesian**)
 - **Deductive** reasoning: Theory is used to develop a set of hypotheses, predict data that would be observed under the theory, and compare the prediction with the actual sample observations (**Frequentist**)
- **Probability methods** play several roles including planning of the study, sample selection and inferential statements

Basic Concepts: Inductive/Deductive

- Consider tossing a fair coin
- We want to know the probability of observing a “head”
- Inductive reasoning: *What would we do?*
- Deductive reasoning: *What would we do?*

Basic Concepts: Inductive/Deductive

- We want to know the probability of observing a “head” (**parameter** = proportion of heads)
- **Inductive:** flip a coin “n” times, count the number of heads (generate sample of data first!)
 - Example: Flip a coin 5 times, 3 heads, $\Pr(\text{“Heads”})=0.6$
 - Example: Flip a coin 1000 times, 499 heads, $\Pr(\text{“Heads”})=0.499$ (infer what the **parameter** is!)
- **Deductive:** Theory implies neither side is more likely. Expect $\Pr(\text{“Heads”})=\Pr(\text{“Tails”})=1/2=0.5$.
 - Now, flip a coin 1000 times, 499 heads.
 - What do I conclude about my theory? Compare the specific data to what the theory predicted/expected.

Design-Based and Model-Based Inference

- **Design-based inference:**
 - Values for survey variables of interest in a clearly defined finite population are **fixed**
 - Inference is based on probability distribution of a sample selection indicator (the **random** variable)
 - Assumes that we have a **probability sample** (or “quasi-randomization”, where we usually estimate probabilities, using a model)
- **Model-based inference:**
 - Values for survey variables of interest are assumed to come from a statistical model (e.g., linear regression model with dependent variable normally distributed)
 - Probability sampling can be used to obtain the sample and taken into account in the modeling.

Design-Based and Frequentist Inference

- Probability of an event, A (flipping a head), is the limit of the relative frequency of the number times the event A occurs among all possible events (flipping a head or tail), in the infinite number of identical replications / trials of the experiment (flipping a coin).
 - To what relative frequency do we converge after infinity trials?
(assuming a fair coin)
- Any “event” has to be a result of an experiment, or we must construct some thought experiment to develop the construct “probability”
- No probability statement can be made if an “event” cannot be construed as a result of a (repeatable) experiment.

Design-Based and Frequentist Inference

- What is a confidence interval in this view?
 - The population parameter, e.g., μ , is **fixed**
 - The interval bounds are **random** (varying across samples), and a function of the sample size
 - Interpretation of a 95% CI: “In 95% of samples I could draw under this specific design, the true mean μ would be within the confidence interval.”
 - An admittedly awkward interpretation for one sample! (Do you agree? Not me)

Design-Based and Frequentist View on Inference

- Design-Based & Frequentist inference uses this view of probability in constructing inferences
- Repeated sampling → many “experiments”, where with a large number of repeated random samples, the distribution of what we are trying to estimate becomes clearer. **Some key concepts:**
 - Unbiased
 - Sampling Variance
 - Mean squared error
 - P-value
 - Consistency

Bayesian View on (Probability) Inference

- Bayesian inference is quite subjective
 - Given **prior beliefs** about the distribution of some parameter of interest, **data**, and a **model** for those data, what is the **updated probability** that the parameter of interest falls within certain values?
 - Two individuals with the same information could arrive at different probabilities for the same event
 - Very subjective (“I have my prior beliefs and my model, and you have yours”)
- Much broader view
 - An experiment could be used to come up with the probability, but it is **not necessary** that the event should be a result of repeated, replicable experiments; (valid inference **(under the model assumptions)** without assuming repeated sampling)
- Any unknown implies uncertainty about the unknown, e.g. model/population parameters like regression coefficients, and hence is subject to a probabilistic statement.
- Prior knowledge about the population parameter of interest is updated by the sample data (not the case for Design-Based and Frequentist methods).

Bayesian Credible Interval

- Once a **model** has been specified, then statements about probability are possible
- For Bayesians, the parameter, e.g., μ , is **random**, and the bounds are **fixed**
- Interpretation of 95% credible interval: “The true value of the parameter, e.g. μ , falls within these bounds with probability 0.95.”

Superpopulation Theory

- In **Design-Based** approach interest is in the finite population parameters (finite population means , variances), which are are fixed.

- The **random selection** of the sample according to a sample design provides the sampling error.

- Inductive inference is used for going from the sample to the target finite population

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- **Superpopulation:** We imagine it as an infinite population (IF) or stochastic process (SP), e.g., an assumed normal distribution, that generates the “fixed” finite population as a finite random sample from IF or SP. In both cases the superpopulation is based on a **model**.

- Under a superpopulation the Census values for a finite population have variance/sampling error, i.e., NOT fixed values.

- Cochran: “...it is far removed from reality to regard the population as a fixed batch of known numbers.”

- Superpopulation model can address variability in estimates due to the finite population being a sample (not fixed) from the superpopulation. E.g., this can account for changes in finite population over time such as occurs for annual surveys.

- Can use a **statistical model (a.k.a. a superpopulation model)** method to account for the additional variability:



 Superpopulation Variability

Summary

- Model-based approaches
 - Bayesian
 - Frequentist
- Design-based (**Randomization Inference**) and Frequentist approaches
 - Adding superpopulation component to the inference adds additional model assumptions.
- Different views on probability basis of the inference.
- These views (can) acknowledge the importance of the sample design

HOW DOES INFERENCE FIT INTO RESEARCH?

Steps in Research Investigations

- What are the research questions and what needs to be estimated and are the hypotheses to be tested?
 - ❖ Often it is a “general purpose” study is designed for many estimates and hypotheses to be tested
- Design (including cross-sectional, longitudinal, case-control [choice-base sample], etc.) and conduct the study/data collection/survey and to obtain sample of observations
- Consider probabilistic frameworks to draw **inference** about the hypotheses based on the data collected (i.e., sample size, degrees of freedom for variance estimation, expected power, sophistication of investigators, etc.)

Research Investigations

- Address the scientific question given the study design
 - Objective is analytic vs. descriptive? Both can use models with parameters, if so, need to formulate statistical models
e.g., analytic linear regression model : $y_i = \beta_0 + x_i\beta_1 + \epsilon_i$;
changing X will change Y or descriptive to estimate the mean of Y for different X 's.
 - The research hypotheses can be formulated as parameters or features of the model (e.g., in a regression is $\beta_1 > 0$)
 - Probabilistic inference about the parameter becomes the goal (Design-based or Model-based? Using modeling in Frequentist or Bayesian? Accounting for Superpopulation in Design based?)

Research Investigations (Contd.)

- ♥ Study features of a **finite population**

- A finite, well-defined population consists of N units
- A survey variable of interest, Y , on these N units are

$$Y_1, Y_2, \dots, Y_N$$

- The goal is to draw inference about some function of these N values,

$$Q(Y) = Q(Y_1, Y_2, \dots, Y_N)$$

- $Q(Y)$ could be a mean, mode, quartile, proportion,..., etc. for the finite population.
- Note that $Q(Y)$ is a computable quantity, if all N values are known (from a Census).

Surveys

- Surveys are a particular type of **research investigation**
- Surveys are often used to estimate **characteristics** of a population
- Surveys are also used for analytically studying associations and relationships often with regression.
- For special studies involving randomized **experiments** can be conducted within surveys
 - *For example, do differences in wording lead to different response distributions and associations?*

Survey Inference

- Survey inference **depends upon the design**
 - **Design-based framework:** what will the sampling distribution look like under repeated sampling following the same randomized design? Cluster sampling, stratification, etc.
 - **Model-based framework:** Is the design ignorable or should it included in the model?
- Let's start with an overview of survey inference
 - Accounting for sample design, nonresponse

Survey Inference

- Draw a sample of size n from the population of size N to obtain the values of Y on the n sampled units.
- Use these values to infer about some quantity of interest $Q(Y)$ (*e.g., a proportion*).
- General principle in drawing sample:
 - **Probability sample:** Every subject/unit in the target population should have a positive and known probability of being included in the sample.
 - **Non-probability sample:** Some/all subjects have an unknown probability of being included in the sample.

Survey Sample Design

- EPSEM: Equal Probability of Selection Method from a population of size N with the same sample size n .
 - Every subject/unit in the population has the same probability of being included in the sample, n/N .
 - Example 1: A simple random sample without replacement (srswor)
 - Every sample of size n from the population has the same chance of being selected. What is the chance? $1/\binom{N}{n}$
How do you get n/N above for srswor? $\binom{N-1}{n-1}/\binom{N}{n} = n/N$.
 - Example 2: For stratified srswor divide the population of N units into H strata with N_h units in stratum $h = 1, 2, \dots, H$. Select a srswor of size $n_h = n \times N_h/N$ from stratum h , i.e., proportional stratified srswor. For each subject/unit in stratum h its probability of selection but $\frac{n_h}{N_h} = \frac{n \times N_h/N}{N_h} = \frac{n}{N} \Rightarrow EPSEM$

Survey Sample Design (Cont.)

- Non-EPSEM designs are common (varying probabilities of selection)
 - Introduces weights vary for sampled units
- Clustering is also very common (especially in the face-to-face household surveys) for reducing the costs, e.g., interviewer travel
 - Example: Two-stage cluster sample
 - Population is divided into clusters
 - A random sample of clusters is first selected, and then units within the sampled clusters are randomly subsampled
 - Multistage cluster sampling is common, e.g., household surveys
- **Complex Sample Design:** Stratification, Cluster Sampling, and Weighting
- **Sampling error:** Error because only a sample is observed instead of the entire population (Census)

Sample selection

- **Target population:** Population for which inference is needed
- **Sampling Frame:** A listing, e.g., addresses or records like birth certificates, used to select the sample from the target population
- Ideally, the sampling frame will include the entire target population, but sometimes it may not
 - This introduces **coverage error**
 - Sampling frame: A telephone survey uses a cellular RDD frame to select a sample of adults (18 years of age or older)
 - Target population: All adults living in the United States
 - Not covered: Those adults who do not have a working cellular phone
 - Assumption: Adults with and without cellular phones are similar on variables of interest (if not, coverage error!)

Nonresponse Error

- Not all units selected will cooperate in providing the survey variables of interest
- This introduces, potentially, a bias in the sample when responding and non-responding units are systematically (not randomly!) different.
- **Assumption:** The differences between respondents and non-respondents can be removed via adjustment using all available variables
- Both coverage and nonresponse error adjustments make unverifiable assumptions → “quasi-randomization” and MAR (missing at random)
- There are other types of errors such as measurement error, processing error, etc. (“Total Survey Error”)
- For now, we focus only on sampling error (that is, uncertainty due to observing a sample rather than the population) and assume there are no other errors.

Design-Based Framework

- Also called “Randomization Inference”
- Focus on randomization
 - Inference accounts for all features of randomization, e.g., sampling probabilities, cluster sampling, etc.

Design- vs. Model-Based Inference

- Design-based inference
 - Seek estimators with good repeated sampling properties, either unbiased or consistent
 - We will review those properties
 - However, it can be shown many estimators have an implicit model
 - For example, ratio or regression estimation
- Limitations
 - Theory starts from probability sampling with 100% response rate
 - Relies on asymptotics → statistical properties, e.g., consistency of the estimators with increasingly large samples

Design-Based Framework

• Basic structure

- Finite population is a collection of N constants, $Y_1, Y_2, \dots, Y_N$
- Estimand of interest: $Q(Y) = Q(Y_1, Y_2, \dots, Y_N)$
 - Examples: population mean, population total, population variance, population regression coefficient, etc.
- Randomness comes from which elements are selected to form a sample
- Define a sampling indicator I_j for $j = 1, 2, \dots, N$:
 $I_j = 1$ if finite population unit j is included in the sample
 $I_j = 0$ otherwise

Sample Design

- Distribution of the sampling indicators
- Start from inclusion probabilities of being in the sample:

$$\pi_j = Pr(I_j = 1)$$

- More generally, this joint probability distribution:

$$Pr(I_1, I_2, \dots, I_N)$$

- These sampling indicators I_j may be correlated

– Examples

- Selection without replacement
- Cluster sampling

Inference

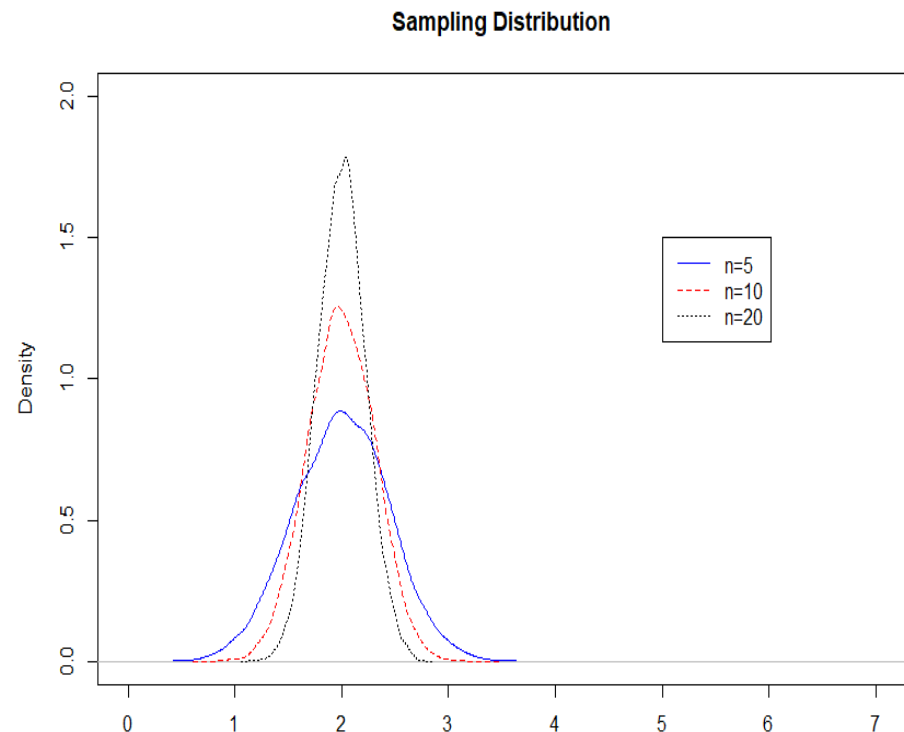
- A procedure is developed for estimating finite population parameter $Q(Y_1, Y_2, \dots, Y_N) = Q(Y)$, which when applied to the sample data gives us an estimate, $q(y_1, y_2, \dots, y_n) = q(y)$.
- One can come up with lots of procedures.
 - For example, taking the case nearest the middle of the sorted list is the estimator of the sample mean
 - Or even nonsensical: take the first five cases from the sorted list, sum their values, add 6, divide by 10, and call that the estimator of the sample mean
- *How do we judge the desirability of these procedures?*

Inference

- Consider the design-based approach first
- Repeated sampling concept:
 - Many samples are possible, as specified by the distribution of the sampling indicator
 - Each sample includes different elements of Y (from finite population) and can give us a different estimate.
 - The many estimates $q(y)$ form a **sampling distribution**.
 - Study the properties of estimates across all possible samples (e.g., what is the standard deviation of the sampling distribution? Sampling variability?)

Sampling Distribution

- Hypothetical **sampling distribution** reflects the distribution of estimates that could be achieved under repeated sampling using the same sample design
- **Example:**
 - $N=100,000$, Population mean=2
 - Draw 1,000 simple random samples
 - Estimate mean of each sample
 - Plot the distribution of these 1,000 means
 - Three different sample sizes:
 - $n=5$
 - $n=10$
 - $n=20$
- In general, larger sample size = smaller sampling error / less sampling variance
- **Design-based inference** based on the shape of sampling distribution drives the inference, e.g., 95%CI, p-value



Design-based Inference

- A repeated sampling procedure is “desirable” if estimates across samples satisfy certain properties
- If the procedure is desirable, then the particular estimate (for the sample in hand), **computed appropriately (with respect to the design)**, is treated as “desirable” and is used for inference
- *What are these “desirable” properties?*
 - Unbiased
 - Minimum (or small) sampling variance
 - Consistency

“Desirable” properties: Unbiased

$Q(Y)$: Population estimand of interest

The set of possible samples are $\{s_1, s_2, \dots, s_S\}$, S (script S) is number of samples.

Let the observation values for sample s_i be denoted by $y^{(i)}$.
(Assume the samples are equally likely to be selected)

The values of the estimator $q(y)$, over the possible sample are $q(y^{(1)}), q(y^{(2)}), \dots, q(y^{(S)})$ then

$$E(q(y)) \equiv \frac{1}{S} \sum_{i=1}^S q(y^{(i)}) = Q(Y)$$

Defines expectation
of $E(q(y))$

Generalized definition of $E(q(s))$:

$p(s_i)$ = Probability of
choosing the sample $s_i, i = 1, 2, \dots, S$

$$E(q(s)) = \sum_{i=1}^S p(s_i) q(y^{(i)}) = Q(Y)$$

“Desirable” properties (Cont.)

- Minimum sampling variance
 - Assume unbiased estimation procedure (often not the case)
 - Choose the sampling procedure that produces the smallest sample to sample variability.
- The variance of the estimator $q(y)$:

$$V(q(y)) = \frac{1}{S} \sum_{i=1}^S (q(y^{(i)}) - E[q(y)])^2$$

Each possible sample is equally likely

$$V(q(y)) = \sum_{i=1}^S p(s_i) (q(y^{(i)}) - E[q(y)])^2$$

Each possible sample s_i has probability $p(s_i)$

- A “desirable” procedure (combining sampling and estimation) is on average close to the target population parameter $Q(Y)$ with the highest possible reliability, i.e., smallest variance.

Bias-Variance Trade-off

- A: (Unbiased + large variance) vs. B: (Small bias + Small variance)
- Two views
 - If unbiased is a must and will be willing to live with large variance the always go with A
 - Willing to consider B, if there is considerable benefit
- Mean Squared Error (MSE) :

$$\text{Bias of } q(y): E(q(y)) - Q(Y)$$

$$\text{Variance of } q(y):$$

$$V(q(y)) = \sum_{i=1}^S p(s_i) (q(y^{(i)}) - E[q(y)])^2$$

$$MSE = Bias^2 + Variance$$

What is Design Consistency?

- The heuristic idea:
 - As the sample size, n , increases, the estimates $q(y)$ should get closer to the target estimand $Q(Y)$.
(allow N target finite population size to also increase)
 - Also, with fixed N as the n increases and gets closer to N , the $q(y)$ should also get closer to $Q(Y)$.
- The stated mathematically:

$$|q(y) - Q(Y)| \rightarrow 0 \text{ as } n \rightarrow N \text{ even if } N \rightarrow \infty$$

Example: Simple Random Sampling Without Replacement (SRSWOR)

- SRSWOR of size n from the finite population $\{Y_1, \dots, Y_N\}$
- $Q(Y) = \bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i$, the Population mean
- Procedure: compute the sample mean, $q(y) = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$
 - Unbiased
 - Minimum sampling variance. Compute the (sampling) variance of $q(y) = \bar{y}$?

$$V(\bar{y}) = \frac{1}{S} \sum_{i=1}^S \left(q(\bar{y}^{(i)}) - Q(Y) \right)^2, \text{ for SRSWOR; recall } \bar{y}^{(i)} \text{ is}$$

sample mean for each sample s_i

$$= (1 - f)S^2/n, \quad f = n/N \text{ the finite population sampling fraction}$$

$$S^2 = \frac{1}{(N-1)} \sum_{i=1}^N (Y_i - Q(Y))^2, \text{ population variance of the } Y_i\text{'s}$$

Sampling Variance Estimation

- Note that the sampling variance for $q(y) = \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ is not known, $V(\bar{y}) = (1 - f)S^2/n$ because S^2 is unknown. Need to estimate S^2 .
- Example for SRSWOR:

Use $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$ to estimate S^2 .

Can show that: $E(s^2) = S^2$, i.e., s^2 is unbiased estimator of S^2 .

Good estimator of $V(\bar{y})$ is $\hat{V}(\bar{y}) = (1 - f)s^2/n$ since $\hat{V}(\bar{y})$ is unbiased estimator of $V(\bar{y})$, i.e., $E[\hat{V}(\bar{y})] = V(\bar{y})$.

Example: Inference for a Superpopulation Mean with SRSWOR

- Let the target finite population values be $\{Y_1, Y_2, \dots, Y_N\}$ be generated from a superpopulation model:
 - Y_i are independent and identically distributed random variables with mean μ and variance σ^2 .
 - The finite population mean is $\bar{Y}$ and the superpopulation mean is μ .
 - Let $\{y_1, y_2, \dots, y_n\}$ be SRSWOR from the finite population with $\bar{y}$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$ the sample mean and sample variance estimators of $\bar{Y}$ and S^2 , respectively.
 - Let E_{MF} and V_{MF} represent the expectation and variance with respect to the superpopulation model/process (M) and respect to to sampling (F) of the finite population, respectively.

Example: Inference for a Superpopulation Mean with SRSWOR(cont.)

- It can be shown that $\bar{y}$ is an unbiased estimator of the superpopulation mean, μ :

$$E_{MF}(\bar{y}) = E_M[E_F(\bar{y}|Y'_i s)] = E_M[\bar{Y}] = \mu.$$

Similarly, it can be shown that $E_{MF}\left(\frac{1}{n}s^2\right) = \frac{1}{n}\sigma^2 = V_{MF}(\bar{y})$, i.e., $\frac{1}{n}s^2$ is an unbiased estimator variance of $\bar{y}$ accounting for both the superpopulation process for obtaining $\{Y_1, Y_2, \dots, Y_N\}$ and the SRSWOR of $\{y_1, y_2, \dots, y_n\}$.

- The unbiased estimator for the variance of $\bar{y}$ (**accounting for only the SRSWOR sampling from the finite population**) is $\left(1 - \frac{n}{N}\right)\frac{1}{n}s^2 < \frac{1}{n}s^2$ the unbiased estimator variance of $\bar{y}$ incorporating both the generation of the finite population from the superpopulation and the SRSWOR.

When we account for the generation of the finite population from the superpopulation, we obtain a larger variance for $\bar{y}$.

Because $E_F\left[\left(1 - \frac{n}{N}\right)\frac{1}{n}s^2\right] < E_{MF}\left(\frac{1}{n}s^2\right)$ implies $V_F(\bar{y}) < V_{MF}(\bar{y})$.

Module 7 Summary

- **Statistical inference:** drawing conclusions about a population based upon a sample/subset of data
- Design-based vs. Model-Based Inference
 - **Design-based inference:**
 - Population values are fixed
 - Inference is based on the sample design, i.e., the probability distribution of sample selection
 - **Model-based inference:**
 - Survey variables realizations/outcomes from a random process defined by a statistical model
 - Inference is based on the selected outcome model

Module 7 Summary

- Design-based viewpoints
 - **Random Survey Samples**
 - Evaluate properties over the “long run” (i.e., theoretical repeated sampling and estimation operation)
 - Select estimators with good “long run” properties
 - **Superpopulation**
 - Can use a model, e.g., Normal Distribution $N(\mu, \sigma^2)$ with mean μ and variance σ^2 , that generates Y values in the finite population $\{Y_1, Y_2, \dots, Y_N\}$. Can be less model dependent; see slide 49. This allows for identifying superpopulation parameters, e.g., μ, σ^2 , that may be of more interest than just the finite population $\bar{Y}, S^2$ and this allows for including extra variability in the inference about $Q(Y) = \bar{Y}$ based on the estimator, e.g., $q(y)$, from the random sample.
- Model-based viewpoints
 - **Bayesian**
 - Probability statements don't rely upon repeated operations
 - Select a (subjective) statistical model for the data including a model for the parameters, which is based on prior beliefs about the parameters.
 - Update (subjective) prior beliefs about the parameters based on observed data.
 - **Frequentist**
 - Probability statements rely upon repeated operations
 - Select a model for generation the data where model has parameters that are not updated by the observed data
 - **Superpopulation**
 - Population parameters are drawn from a distribution of possible parameters
 - Select a model to represent the superpopulation

Module 7 Summary

- Sample surveys are a particular process for gathering data
- Specific approaches to doing inference are needed
- We will continue our discussion of randomization (design-based) inference next week, and then turn to model-based inference